

2.5 MW Pulse Power Supply

Team Number: 57

Team Name: Pulse Power Innovators

Team Members:

Nicholas Bockrath

Sean Reeves

Diyorbek Egamov

Nicholas Pelino

Project Mentor: Jesus Alejandro Valladares (Los Alamos National Laboratory)

Point of Contact: chuy@lanl.gov

Class: EEE 489, Spring Semester

Date: February 6, 2026

Table of Contents

Executive Summary	2
Introduction	2
Background	2
Roadmap	3
Completed Tasks	3
Project Scope	4
Final Demonstration Day Vision	4
Budget Report	4
Changes to Project Scope	5
Schedule	5
Completed Tasks to Date	6
In-Progress Tasks	7
Impact of Delays	8
Team Specifications and Background	8
ABET Considerations	9
Conclusion	10
References	10

List of Figures/Tables

Figure 1	7
Figure 2	8
Figure 3	8

Executive Summary

The goal of this project is to build a pulsed-width modulator prototype that is safe, but realistic, enough to be used in an academic environment in order to fill the lack of hands-on education on PWMs in the college laboratory. [What you've built so far]. [What remains].

Introduction

Pulsed power systems deliver energy in short, high-intensity bursts, enabling very high instantaneous power from modest average sources. They are essential to applications such as particle accelerators, radar, medical devices, and advanced industrial systems. However, full-scale pulsed-power modulators are unsafe and impractical for student laboratories, creating a gap between academic theory and real-world systems.

A low-voltage educational prototype addresses this gap by safely demonstrating pulse formation, energy transfer, and waveform control. This hands-on platform connects theory to measurable behavior while preparing students for

careers in high-reliability research and national laboratory environments, and it is the main objective of this project.

Background

Pulsed power systems are widely used in modern electrical and industrial applications where high peak power is required over short durations. Unlike conventional continuous power supplies, pulsed power systems accumulate energy over time and release it rapidly, enabling high instantaneous power while maintaining relatively low average power consumption. This principle is fundamental to applications such as particle accelerators, pulsed radar systems, medical devices (e.g., defibrillators), and high-energy physics experiments [1], [2].

Several established pulsed power topologies exist in industry, each suited for specific voltage, current, and timing requirements. One of the most famous is the Marx generator, which charges capacitors in parallel and discharges them in series to achieve high voltage outputs [2]. Another common configuration is the Blumlein pulse forming network (PFN), which provides rectangular pulses with fast rise times and controlled impedance characteristics [2], [3], [4].

The system proposed in this project builds upon established pulsed power concepts while focusing on a low-voltage, microcontroller-controlled implementation suitable for educational use. The use of adjustable PWM control, current limiting, and real-time waveform visualization provides an educational platform for demonstrating switching behavior.

The novelty of this project lies in its combination of accessibility and functionality, offering a system that is both safe for laboratory environments and representative of real-world pulsed power systems. By incorporating both hardware and software elements, the design lets users explore the interaction between control algorithms and power electronics, providing valuable insight into modern electrical engineering practices.

Roadmap

This report first gives a summary of what the team achieved last semester in the Completed Tasks Section. Then, it explains what the goals of this project are via sections like the Project Scope, Budget, and Final Demonstration Day Vision, which explain what the final design is, the budget and bill of materials, and what the Demo Day will consist of. This report then lays out how the team plans on achieving said goals via the Schedule, listing out each task to be done week by week. Finally, the project will conclude by discussing the ABET Considerations.

Completed Tasks

In the previous semester, the team primarily focused on researching pulsed power topologies and creating a circuit schematic that fits the team's goals and budget. There are various different applications in industry that require different parameters in voltage levels or pulse width, which will inherently change the circuit components and connections. The team created a schematic inspired by the single stage pulse generator topology that fits within the scope of the project. More research was conducted to determine the topology is solid state, which resulted in the inclusion of a gate driver that controls the MOSFETs switches. Included in the circuit as well is a STM32 series microcontroller that will manage the pulse width modulation, which must be programmed in an IDE. The team has familiarized themselves with STM32CubeIDE, a software that uses C as the predominant language for program logic, which will be crucial for controlling the embedded systems in the physical build.

Project Scope

This project presents the design, construction, and validation of an adjustable pulsed DC power supply using PWM switching techniques. The hardware features N-channel power MOSFETs, isolated gate drivers, current-sense amplifiers, and microcontroller-based control enabling user-adjustable frequency, duty cycle, voltage, and current limiting. Physical implementation addresses PCB layout for high-frequency switching, thermal management, and EMI mitigation. Experimental validation includes oscilloscope waveform characterization, efficiency analysis, and current-limiting verification, demonstrating practical tradeoffs in switching converter design suitable for motor control, LED driving, and battery charging applications.

Final Demonstration Day Vision

The presentation begins by introducing the project objectives and final specifications (100Hz-10kHz frequency, 1-99% duty cycle, 1-48V output, 0.1-10A current limiting), then provides a hardware walkthrough identifying MOSFETs, gate drivers, current sensor, microcontroller, and explaining the switching topology and PCB layout considerations. The operating demonstration displays waveforms at various frequencies on the oscilloscope, shows voltage output at different duty cycles, demonstrates current limiting under overload, and presents timing measurements. Performance results include efficiency data across the operating range, thermal behavior under sustained operation, and comparison of measured versus theoretical values. A practical application demo shows LED dimming or motor speed control, followed by a brief conclusion summarizing achieved performance

and design challenges overcome. The session concludes with Q&A where evaluators ask technical questions about design decisions and circuit behavior.

Budget Report

Total Estimated Project Cost: \$185.00

Category	Cost
Power Components (MOSFETs, gate drivers, capacitors, heatsinks)	\$68.00
Control Electronics (microcontroller, sensors, interface components)	\$42.00
PCB & Assembly Materials (board, terminals, wire, solder)	\$35.00
Connectors & Hardware (jacks, enclosure, mounting)	\$18.00
Testing & Documentation (load resistors, thermal paste, printing)	\$22.00
Required Lab Equipment (Institution-Provided): Oscilloscope, multimeter, DC power supply (0-50V, 5A), soldering station	

Changes to Project Scope

The project scope largely remains the same as last semester, using a lower voltage pulsed power model to demonstrate the behavior of a large scale modulation system. Throughout the design process last semester, the team found the scope to be achievable while challenging which resulted in no major changes to the project objectives.

Schedule

Phase 1: Design & Initial Planning (Week 1, Feb 6 - Feb 12) Finalize circuit schematic, component specifications, and Bill of Materials. Place component orders.
Deliverable: First Written Report (Project Synopsis) - Feb 6

Phase 2: Progress Documentation & Procurement (Weeks 2-3, Feb 13 - Feb 26)
Receive and inventory components. Breadboard-test gate driver and current sensing circuits. Deliverable: Video Progress Report - Feb 16

Phase 3: PCB Assembly & Initial Testing (Weeks 4-7, Feb 27 - Mar 26) Solder all components including power stage, control circuitry, and user interface. Perform continuity testing. Power up system and verify voltage rails. Milestone: TA Conference - Mar 23-Apr 1

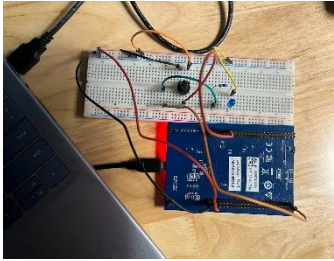
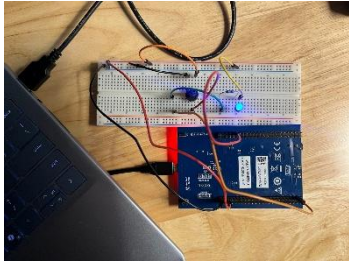
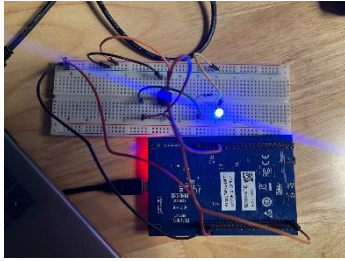
Phase 4: Firmware Development & Debugging (Weeks 8-9, Mar 27 - Apr 9) Write microcontroller code for PWM generation, user interface, and current limiting. Debug switching behavior and test parameter ranges. Deliverables: Second Written Report (Interim Report) - Apr 7, Capstone Showcase Checklist - Apr 9

Phase 5: Performance Testing & Documentation (Weeks 10-11, Apr 10 - Apr 23) Measure switching waveforms, efficiency, and thermal performance. Test current limiting and capture data. Begin final report and prepare demo setup. Deliverables: Project/Team Assessment - Apr 17, Demo Day Intake Form - Apr 20, Senior Design Prize Submission - Apr 22

Phase 6: Final Demo & Submission (Week 12, Apr 24 - Apr 29) Complete presentation slides and poster. Rehearse demonstration. Finalize technical report with all experimental results. Deliverables: Final Report - Apr 29, Course Evaluation - Apr 29

Completed Tasks to Date

The main task completed at this point in the semester is the STM32 microcontroller programming that controls the pulse width modulation. Controlling the PWM is an essential part of the project as it allows the user to change the duty cycle in real time, directly changing the average output voltage delivered to the load. The duty cycle is changed through a potentiometer and its current magnitude is represented with a blue LED for visual simplicity. This involved programming the STM32F407-Discovery board inside of its native software STM32CubeIDE by mapping the desired inputs/outputs for circuit components and implementing the correct functions in user code sections. As the team receives the rest of the ordered parts, the PWM will be implemented with the gate driver that controls the switching of the MOSFET and drives the actual pulsing mechanism.

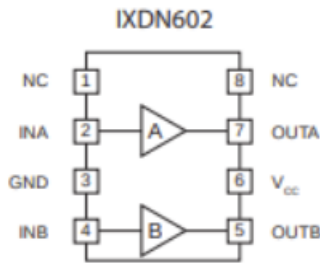
		
LED duty cycle = 0%	LED duty cycle = 50%	LED duty cycle = 100%

[1] Microcontroller controlling blue LED duty cycle at various percentages in real time

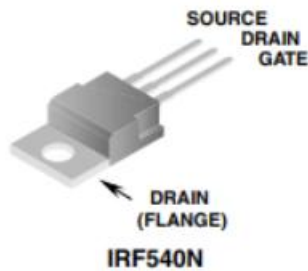
The circuit shown above is simple and emits a constant voltage stream whose magnitude can be varied through the user input. When the final circuit is created, the LED will switch on and off at the programmed frequency instead of being constantly lit, while the duty cycle is still able to be varied through the potentiometer. This allows the team to demonstrate the overall concept of pulse power and changing the parameters results in a much more visible difference in output.

In-Progress Tasks

The primary goal for the team this semester is to take the power circuit schematic created the previous semester and physically construct the circuit to be ready for the final demonstration. To achieve this, the team must test the design and ensure the values of important pulse power variables such as voltage, current, pulse width, pulse frequency, and rise time. All of these values can be tested using the Tektronix DPO-4032 oscilloscope provided by the laboratory. Adapting is an important part of the testing process as well, as changes may need to be made if the values are wrong or the circuit does not behave as expected. With the main microcontroller section done, the task of implementing the MOSFET and gate driver would be the natural next step. The team is currently waiting for a delivery to assemble this crucial part of the topology, but it should be a rather quick and simple execution when the gate driver gets delivered. In the meantime, an LCD screen is currently being integrated to showcase the duty cycle in % as the user changes the potentiometer, as another means of visual aid during the presentation. After testing and the desired values are reached, components must be assembled for demonstration day and presented in a manner that is easily understood by the viewer. The team intends to do this by showing oscilloscope output for pulse width / voltage and LED activation during the pulse for visual aid.



[2] Pin configuration of IXDN602 gate driver



[3] Pin configuration of IRF540N MOSFET

Impact of delays

This capstone project is being completed by a hybrid team, consisting of both online and on campus students. In an ideal scenario, software simulations will be done by online students and the hardware part will be built by two on campus students. However, there is a potential delay, considering shipments or rebuilding of the prototype might take longer than expected in case of failure in the first trials or need for replacing a part. To minimize the risk, our team will make sure to complete all the planning and simulation, much before the Demo day and online orders will be placed accordingly. The team does not anticipate the need to order anymore parts once the current components are delivered. Because a delivery is being waited on, the pulse generation of the project is delayed but remains on schedule to be completed before demo day.

Team Capabilities and Specifications

Sean Reeves -

contributes to the team's electrical engineering efforts with a background in control systems, analog electronics, and circuit analysis. His experience includes working with feedback systems, semiconductor fundamentals, and modeling tools such as MATLAB, LTspice and Simulink for system analysis and validation.

He supports both the analytical and practical aspects of the project, assisting with system modeling, circuit design, and interpretation of simulation results. He also contributes to hands-on implementation, including breadboarding and troubleshooting, helping connect theoretical design with physical systems.

Within the team, he collaborates on technical decisions and provides structured analysis to support overall system performance and reliability.

Nicholas Bockrath - contributes to the team's electrical engineering efforts with a background in control systems, analog electronics, circuit analysis, and signal systems. His experience includes working with feedback systems, semiconductor fundamentals, powerplant design, distribution, and modeling tools such as MATLAB, LTSpice and Simulink for system analysis and validation.

He is the team lead, managing all that the team does, keeping the project mentor updated, and documenting everything that happens as the project progresses.

Nick Pelino -

contributes to the team's electrical engineering efforts with a developing academic background in circuit analysis, control systems, and basic electronics. His coursework has provided exposure to system modeling concepts and analytical problem solving.

He supports the team by assisting with calculations, simulations, and interpretation of results, while continuing to build experience in applying theoretical concepts to practical designs. He collaborates with team members to ensure a clear understanding of system behavior and contributes to overall project development.

Diyorbek Egamov -

contributes to the team's electrical engineering work through his academic studies in circuit fundamentals and introductory system analysis. His coursework has focused on building a strong foundation in electrical principles and problem-solving techniques.

He assists with analytical tasks, including basic modeling, calculations, and reviewing system behavior, while gaining experience in applying classroom knowledge to project-based work. He works alongside the team to support design understanding and contribute to project progress.

GWC 297 Room Facilities –

Hardware provided: digital multimeter, oscilloscope, connection wires, capacitors (many units), resistors, cutting tools, breadboards, and many more mini tools for testing and circuit building.

Softwares provided and used: Waveform Generator, Microsoft Excel (for recording values), LTSpice, Diligent platform, multimeter readings

ABET Considerations

Sustainability: Pulsed power technology is inherently linked to sustainability in the form of its energy usage. The alternative to supplying a quick pulse of voltage is through conventional DC or AC methods, which may not be necessary for all applications. Controlled bursts will **reduce power consumption**, which can be particularly useful in machinery that require extremely high-voltage levels to run, such as a particle accelerator.

Manufacturability: Given the constraints of the supplies allocated to the group, the pulsed power display will be **easier to manufacture** when compared to industrial applications. During the design process, the team is focused on making an affordable demonstration without sacrificing quality of results. This pulse power supply is not meant for high-voltage but to demonstrate to a customer how this method works and how it can be implemented in certain machinery. The team's goal is to design a power supply not only easily comprehensible to a customer, but one that is easily reproducible as well.

Safety: The specific low-voltage topology in this project was chosen with the idea of customer safety in mind. Pulsed power modulation is much more common in high-voltage applications, but due to **safety concerns** commonly present in such situations, a low-voltage model only for demonstration will be used instead. The design will also use devices previously mentioned to ensure safety while testing, such as bleeder resistors and TVS diodes.

Conclusion

Team Pulse Power Innovators has chosen an important project that can prototype a safe high-power supply for the public use. Initial software simulations and complete roadmap for this semester already completed. Hardware parts are being shipped, and the team is close to finalizing the entire project. By the end of the semester, the team intends to showcase a fully-functioning prototype of a Pulse Power Supply.

References

- [1] J. C. Martin, *Pulsed Power Engineering*. Boca Raton, FL, USA: CRC Press, 2015.
- [2] H. Bluhm, *Pulsed Power Systems: Principles and Applications*. Berlin, Germany: Springer, 2006.
- [3] M. Reiser, *Theory and Design of Charged Particle Beams*, 2nd ed. Hoboken, NJ, USA: Wiley-VCH, 2008.
- [4] A. Greenwood, "Solid-state switching in pulsed power applications," *IEEE Trans. Plasma Sci.*, vol. 45, no. 6, pp. 1234–1242, Jun. 2017.